

Computing descriptive statistics of ADHD medication using the OpenSAFELY platform

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Official

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Why use OpenSAFELY?



Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

ADHD and Covid 19 pandemic

The Covid 19 pandemic has affected the provision of mental health services including ADHD services such as:

- a) Diagnosis
- b) Medication

Historically there has been no specific data publicly available about people with ADHD.

Previously statistics on ADHD medication pre and post Covid, such as on Figure 1, rely on rudimentary assumptions.

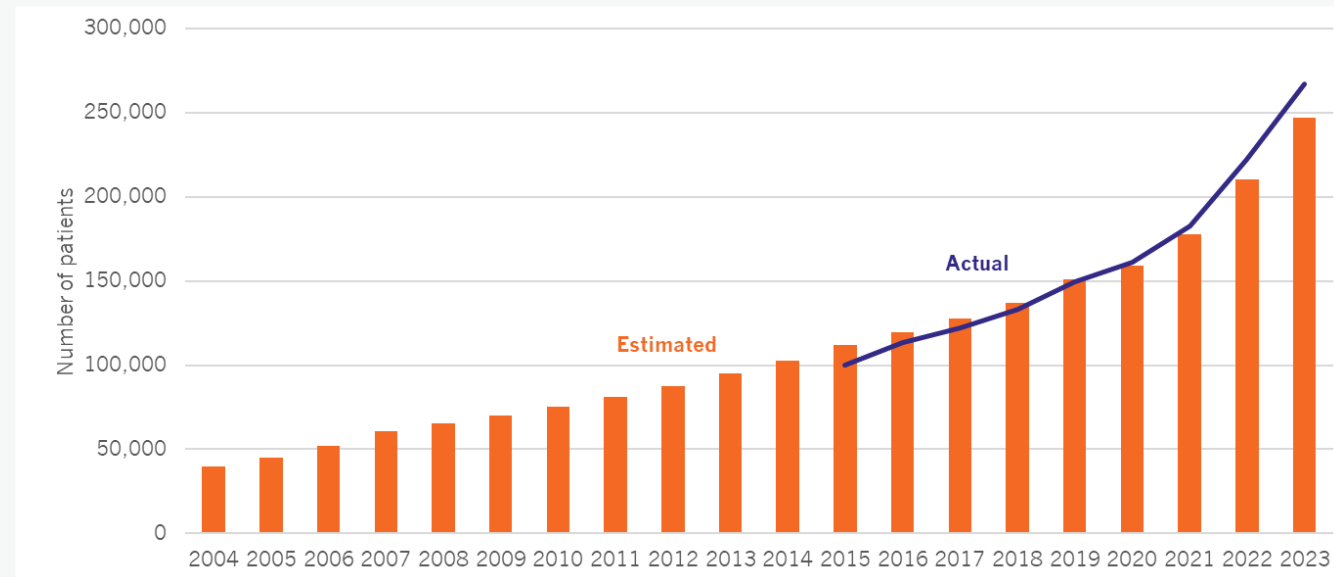


Figure 1: The estimated number of patients being prescribed medication for ADHD, taken from [Lord Darzi's report on the state of the National Health Service in England](#). The figure uses the rudimentary assumption of each patient uses 11.5 items per year based on observations.

NHS England's work on ADHD Data

OpenSAFELY allows medication records to join with event including diagnostics records to provide accurate and precise statistics.

Our statistics will contribute to the [National Level Management information for ADHD](#)

This is a series of work by NHS England to address the challenges of ADHD care – part of the workstream includes the [ADHD Taskforce](#).

Ongoing work to improve care

NHS England will work with colleagues across government to assess and respond to the Taskforce's final report after it is published.

In addition and alongside this, NHS England's ADHD Improvement Programme has been delivering on the commitments made in spring 2024, including:

- Transforming ADHD data to improve data quality and consistency, through:
 - Publishing the [ADHD data improvement plan](#) (May 2025) and corresponding [technical guidance](#) (June 2025)
 - Creation of the new [Neurodevelopmental Data Hub](#), a central repository for neurodevelopmental data produced by NHS England
 - Launching the first ever publication of [national level management information](#) in relation to ADHD – a quarterly publication which began in May 2025

Figure 1: A screenshot from [NHS England](#) on a series of ongoing work for ADHD data transformation

OpenSAFELY

OpenSAFELY is a trusted research environment that allows access to GP data through The Phoenix Partnership's SystemOne

OpenSAFELY allows users to query aggregated records without viewing individual patients

We have created a [GitHub repo](#) and use ehrQL (Figure 1) to query records for patients with ADHD pre and post Covid.

```
dataset.first_mph_med_date = (  
    medications.where(True)  
    .where(medications.dmd_code.is_in(adhd_medication_codelist))  
    .where(medications.date.is_on_or_after(dataset.first_adhd_diagnosis_date))  
    .sort_by(medications.date)  
    .first_for_patient()  
    .date  
)
```

Figure 1: Screenshot of part of an [ehrQL query](#) to capture the first date that patients took their ADHD medication

Workflow of OpenSAFELY

OpenSAFELY discourages developing code directly on primary care data – they provide fake data.

The platform has two workflow:

1. DEV – using fake data to develop code
2. LIVE – using the developed code query primary care data

OpenSAFELY will assign you an inhouse analyst “Copilot” to assist with development

- Full time for the first 4 weeks
- Ad hoc thereafter

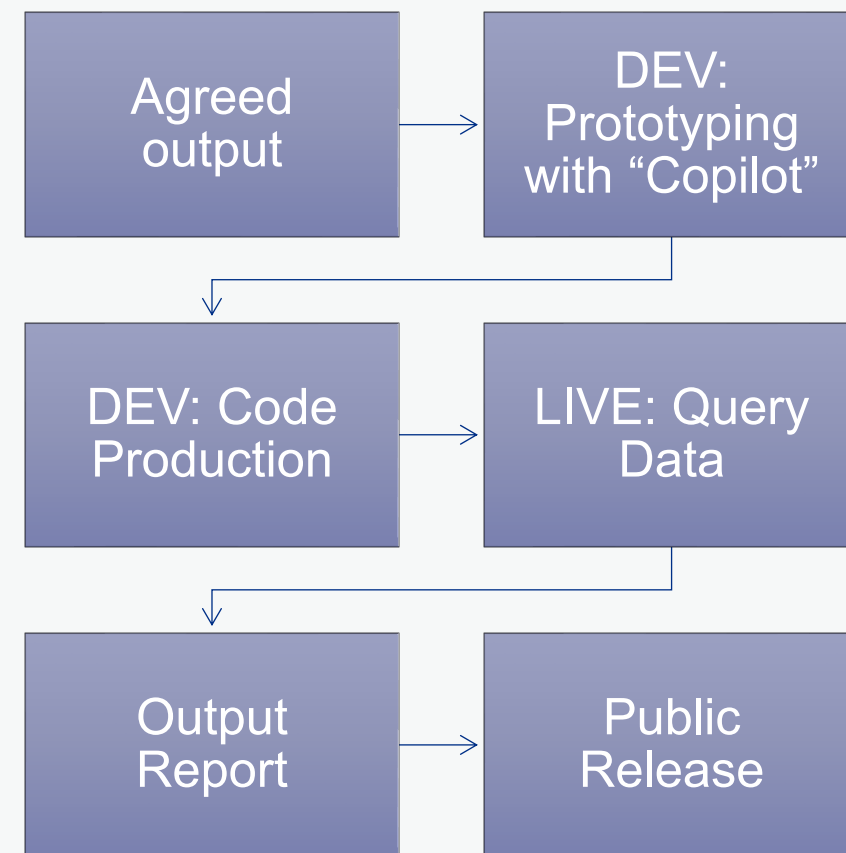


Figure 1: A flow chart of the key steps from creating the repo to public release

ADHD Pathway and Data Architecture

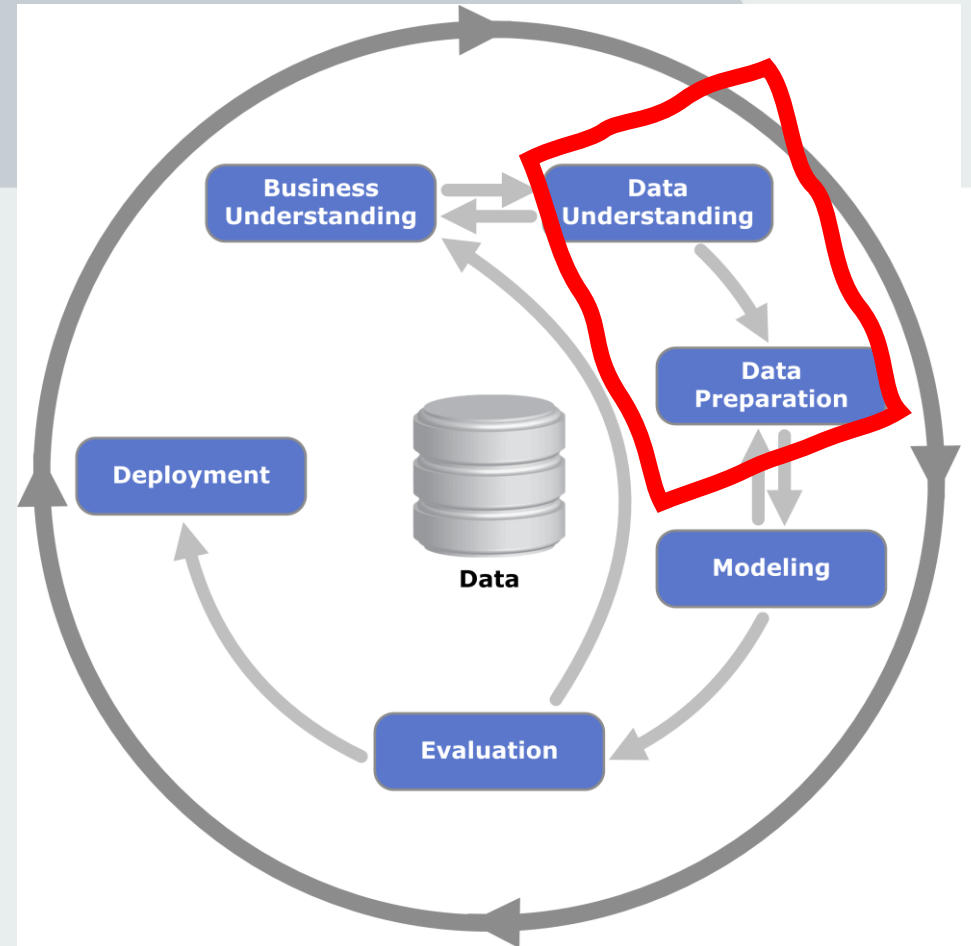


Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

Patient Pathway

We defined the patient pathway by grouping key events displayed on Figure 1.

We analyse how the data architecture OpenSAFELY aligns with the ADHD pathway

For our work, we are not looking at nonpharmacological treatment. There are open questions such as:

- Which SNOMED code to analyse
- If the SNOMED code are used

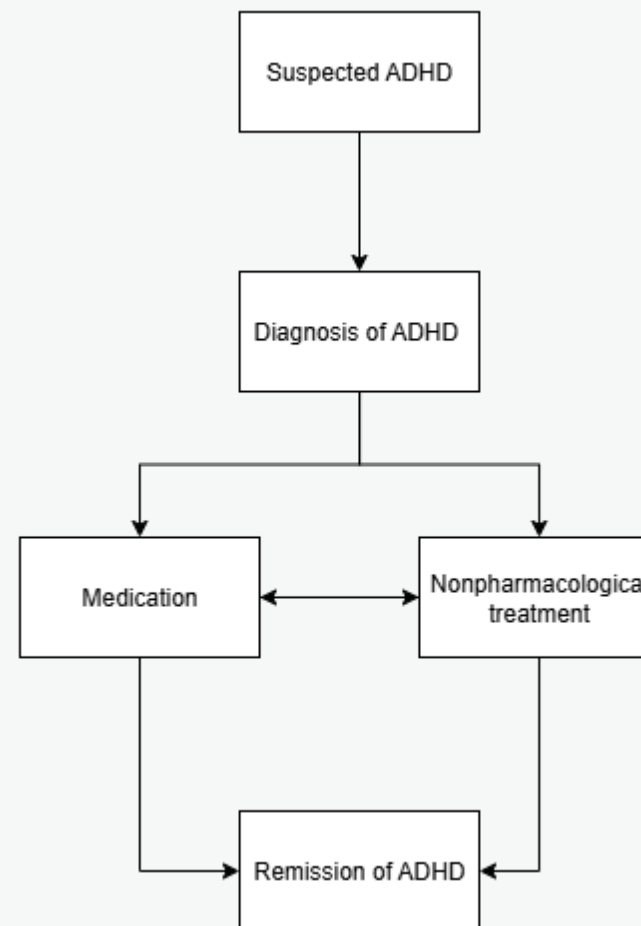


Figure 1: A flow chart on the patient pathway for ADHD in primary care

ADHD Suspected SNOMED Code

Suspected ADHD SNOMED Codes only existed since late 2023

Due to OpenSAFELY’s Covid-related focus, there is no pre-Covid baseline for comparison.

This precludes any information at the beginning of the patient's pathway

Release: uk-edition v20250730

Perspective: Full

About

Leading healthcare terminology worldwide

Concept Details

Concept Details

SummaryDetailsDiagramExpressionRefsetsMembersReferences

Classification Map

Attention deficit hyperactivity disorder suspected (situation)

📄 ⓘ

SCTID: 1863681000000108 , Fully defined , Active

National Health Service realm language reference set (clinical part)

Term	Acceptability (GB)
F ☆ Attention deficit hyperactivity disorder suspected (situation)	Preferred ⓘ

DescriptionId	Type	Language	Case Significance	Effective Time	ModuleId
3461111000000113	Synonym	en	Entire term case insensitive	20231122	999000011000000103

S ✓	ADHD (attention deficit hyperactivity disorder) suspected	Acceptable ⓘ
S ✓	Attention deficit hyperactivity disorder suspected	Acceptable ⓘ
S ✓	Suspected ADHD (attention deficit hyperactivity disorder)	Acceptable ⓘ

Figure: [Screenshot from NHS SNOMED CT browser](#) showing the effective date of suspected ADHD codes (highlighted in yellow)

ADHD Remission SNOMED Code

When looking at [SNOMED usage](#), we find fewer than 10 SNOMED usage for remission of ADHD in 2024.

It is [NHS England's position](#) that true remission of ADHD is rare.

This precludes any information at the end of the patients' pathway.

SNOMED_Concept_ID	Description	Usage
698692009	Attention deficit hyperactivity disorder, predominantly hyperactive impulsive type in remission (disorder)	*

Figure 1: Screenshot from [analysis](#) showing fewer than 10 patients have ADHD remission code in 2024

Querying Code Usage

Thus, the remaining steps we can analyse are diagnosis and medication of ADHD.

We defined the [SNOMED and dm+d codes for diagnosis and medication](#) respectively from inhouse clinicians.

From the data structure on Figure 1, we defined the business rules selecting the date used to calculate prevalence.

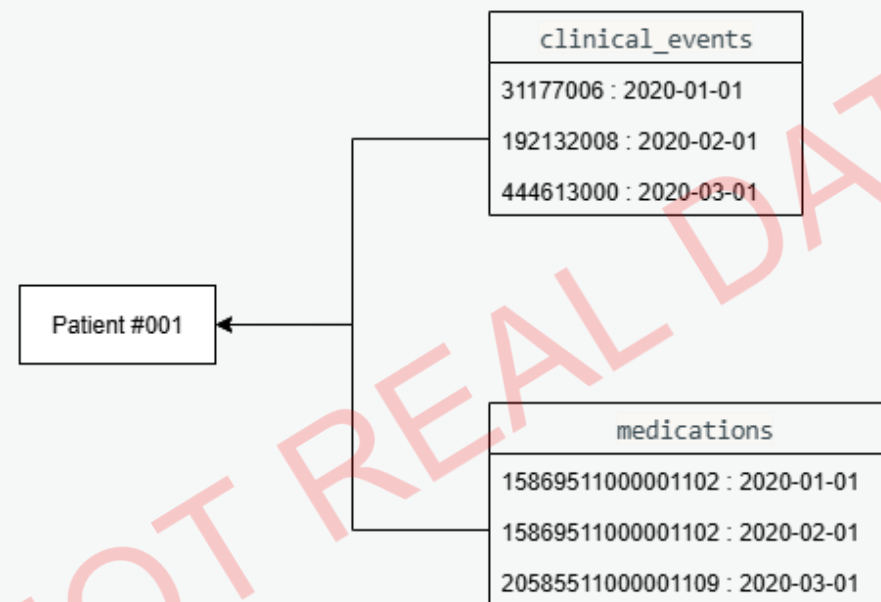


Figure 1: A schematic diagram of the data structure for each patient on TPP. Each patient is linked with a table of clinical events and medication, listing SNOMED CT and dm+d codes respectively with dates.

Comparing Data with other Providers

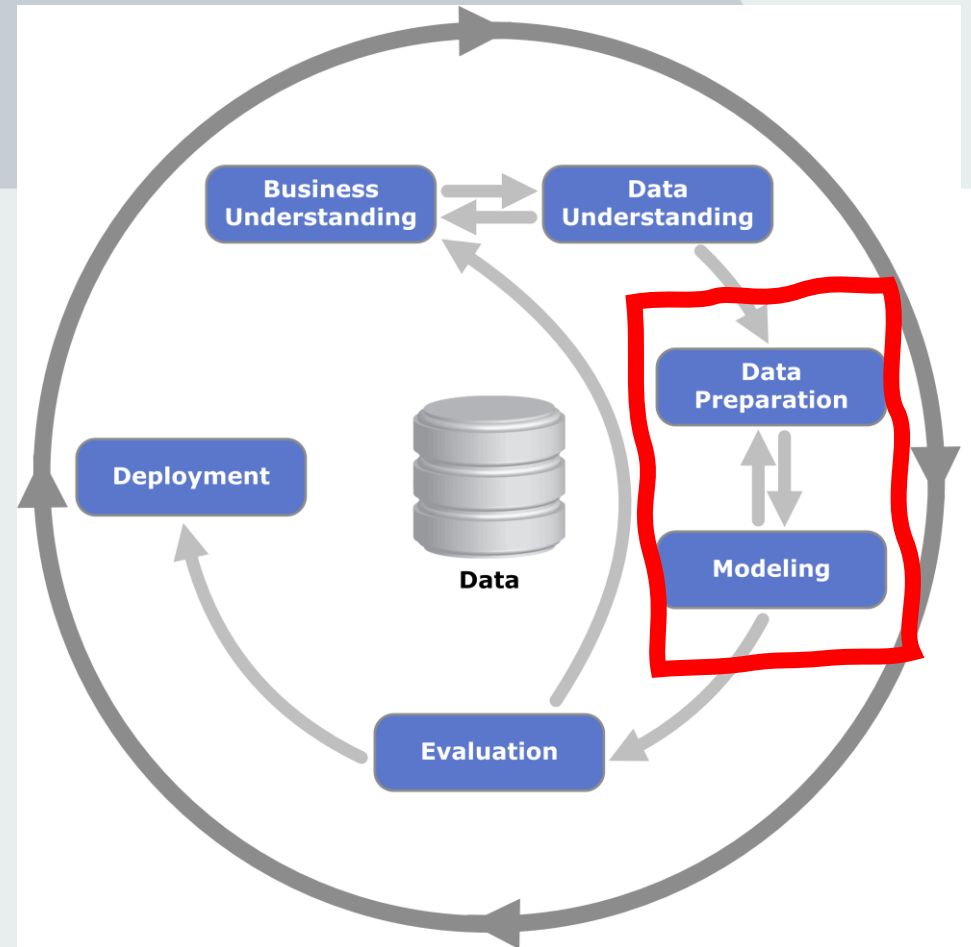


Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

Modelling the ADHD population

We want to compare data in OpenSAFELY based on TPP to other providers. Thus, showing that the statistics are representative of the entire population (in England).

From [NHS England’s Health and Care of People with Learning Disabilities publication series](#) we can calculate the counts of patients with an ADHD diagnosis from EMIS and Cegedim.

- We can compare our statistics from OpenSAFELY with the published data to quantify the similarity of dataset across different providers.

Table						
interval_start	interval_end	ratio	numerator	denominator	sex	age_band
2023-04-01	2024-03-31	0.00320342	5845	1824615	female	0 to 9
2023-04-01	2024-03-31	0.018973	31560	1663420	female	10 to 17
2023-04-01	2024-03-31	0.022808	34620	1517890	female	18 to 24
2023-04-01	2024-03-31	0.0165178	43650	2642605	female	25 to 34
2023-04-01	2024-03-31	0.0092484	23415	2531790	female	35 to 44
2023-04-01	2024-03-31	0.00520493	10995	2112420	female	45 to 54
2023-04-01	2024-03-31	0.00205202	4085	1990720	female	55 to 64
2023-04-01	2024-03-31	0.000733802	1045	1424090	female	65 to 74
2023-04-01	2024-03-31	0.000312235	475	1521290	female	75 and over
2023-04-01	2024-03-31	0.0091562	17555	1917280	male	0 to 9
2023-04-01	2024-03-31	0.0486443	84965	1746660	male	10 to 17
2023-04-01	2024-03-31	0.0398677	59235	1485790	male	18 to 24
2023-04-01	2024-03-31	0.0262691	70220	2673100	male	25 to 34
2023-04-01	2024-03-31	0.0111825	29990	2681860	male	35 to 44

Figure 1: A sample [table](#) showing the counts of ADHD diagnosed patients from EMIS + Cegedim

Calculating ADHD Diagnosis

We follow the same business rules from NHS E’s Learning Disabilities Data:

- 1. For each financial year, we consider patients that were registered for a least one day on a TPP provider.
- 2. We choose the last diagnosis date.

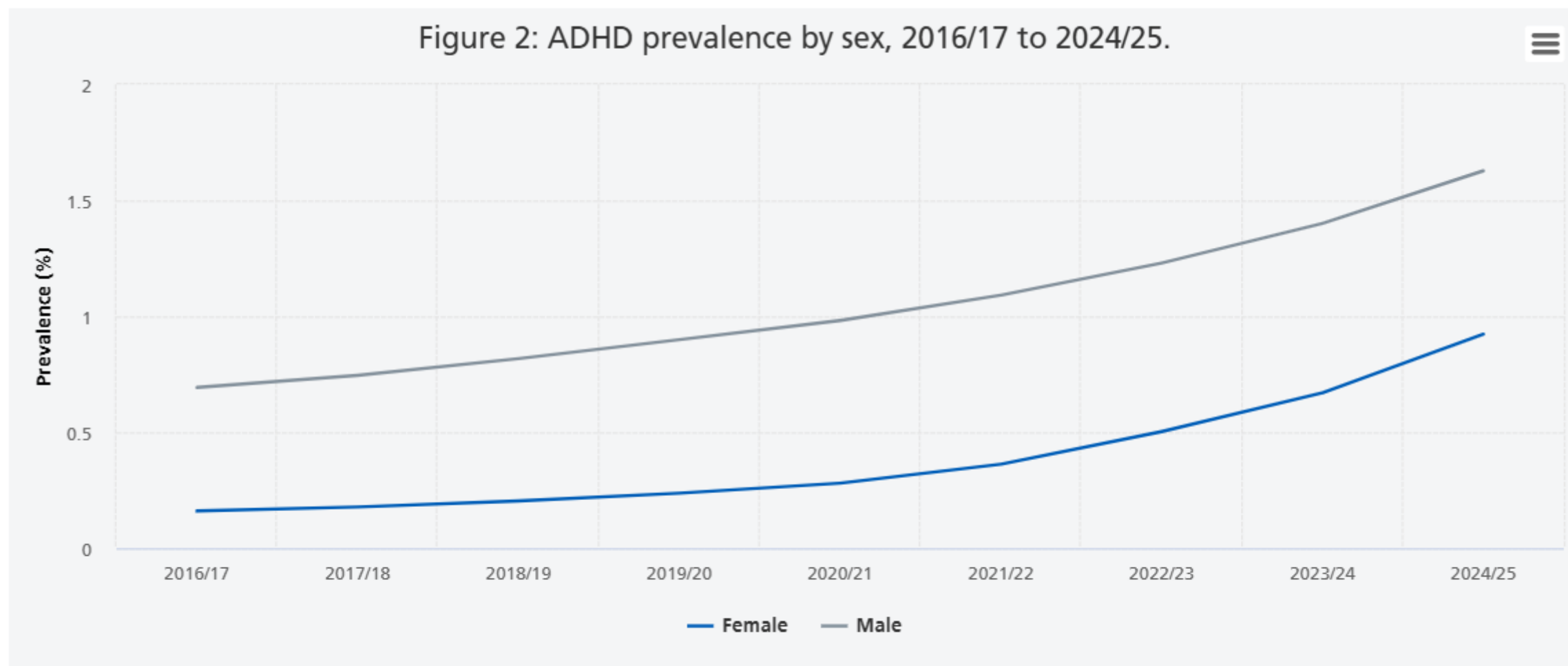
Consider the following case studies:

- 1. Laura was last diagnosed in 2012 and has been registered TPP ever since.
- 2. Helen was last diagnosed in 2017 in TPP then moved to EMIS in August 2021 and returned to TPP in January 2024.
- 3. Douglas was last diagnosed in 2019 and remained in TPP until his death in 2022

Year	Laura	Helen	Douglas
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

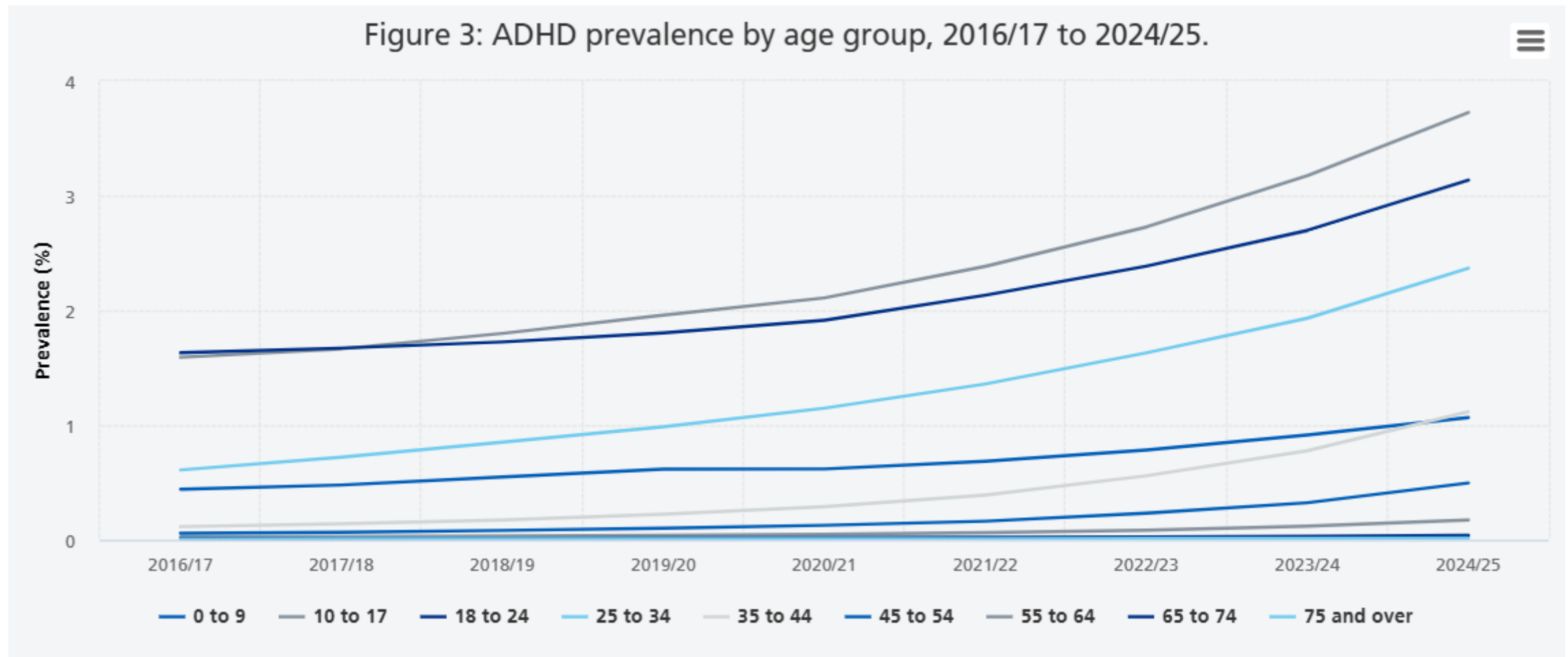
Figure 1: A table showing how each case study would be counted (highlighted in blue) for each financial year.

Result: Diagnosis by Sex



[Download the data for this chart Figure 2: ADHD prevalence by sex, 2016/17 to 2024/25.](#)

Result: Diagnosis by Age Band



[Download the data for this chart Figure 3: ADHD prevalence by age group, 2016/17 to 2024/25.](#)

Comparing ADHD Prevalence

Figure 1 compares OpenSAFELY TPP data and NHS E's published EMIS and Cegedim data.

Each data point is calculated by each year, age band and sex combination:

$$\text{Prevalence} = \frac{\text{Counts of patients with an ADHD diagnosis code}}{\text{Counts of system registered patients}}$$

The recorded prevalence of ADHD was found to be consistent across both primary care datasets.

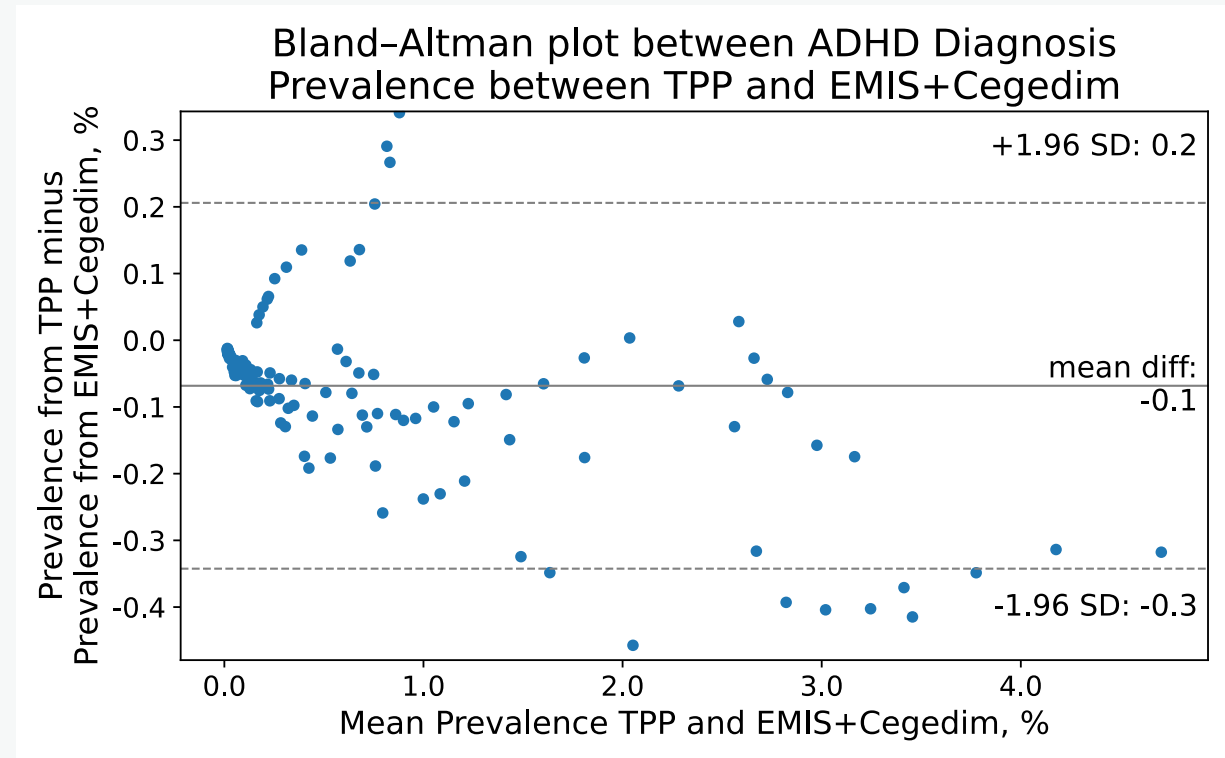


Figure 1: Bland-Altman plot comparing prevalence of ADHD diagnosis between OpenSAFELY TPP data and NHS E's published EMIS and Cegedim data. Each point is broken by both age band, sex and year.

Defining Business Rules for Medication

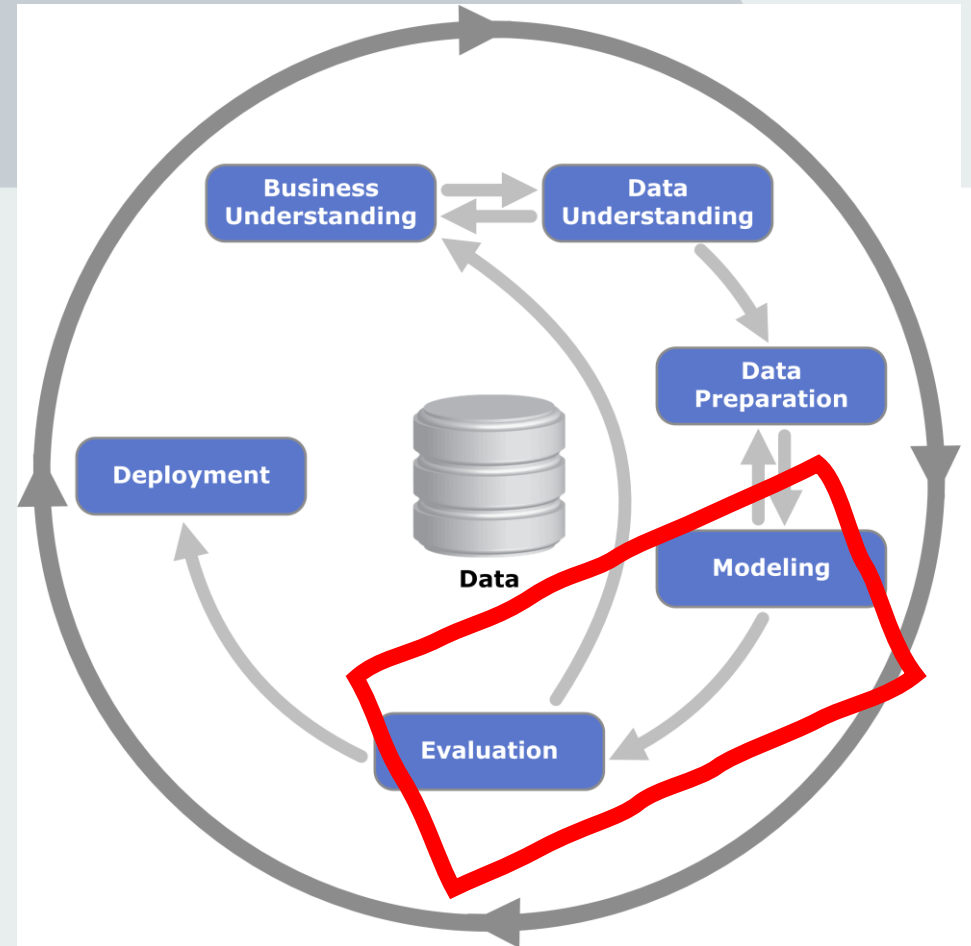


Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

ADHD Medication

We consider the following cuts for ADHD medication:

- Long term perspective – patients have taken ADHD medication for their ADHD at least once in their life.
 - We consider the first ADHD diagnosis and medication.
- Short term perspective – patients that have taken ADHD medication in the last 6 months

For the dm+d medication code, we request prescription only drugs for ADHD.

dm+d codes	Term
12140311000001102	Methylphenidate 20mg/5ml oral solution
37403811000001105	Atomoxetine 25mg capsules
42640611000001108	Lisdexamfetamine 60mg capsules

Figure: A sample of the dm+d [codes](#) used to determine if a patient is prescribed ADHD medication.

Medication Caveat – Linking with Diagnosis Codes

Several ADHD medication are used for different conditions. For example, Methylphenidate is used for [narcolepsy](#).

There is no metadata in OpenSAFELY that links medications to the diagnosis of the condition. Figure 1 shows a case where it is indeterminable to match Methylphenidate to the prescribed condition.

Date	Jul-24	Aug-24	Sep-24
ADHD Diagnosis Code			
Methylphenidate Medication Code			
Narcolepsy Diagnosis Code			

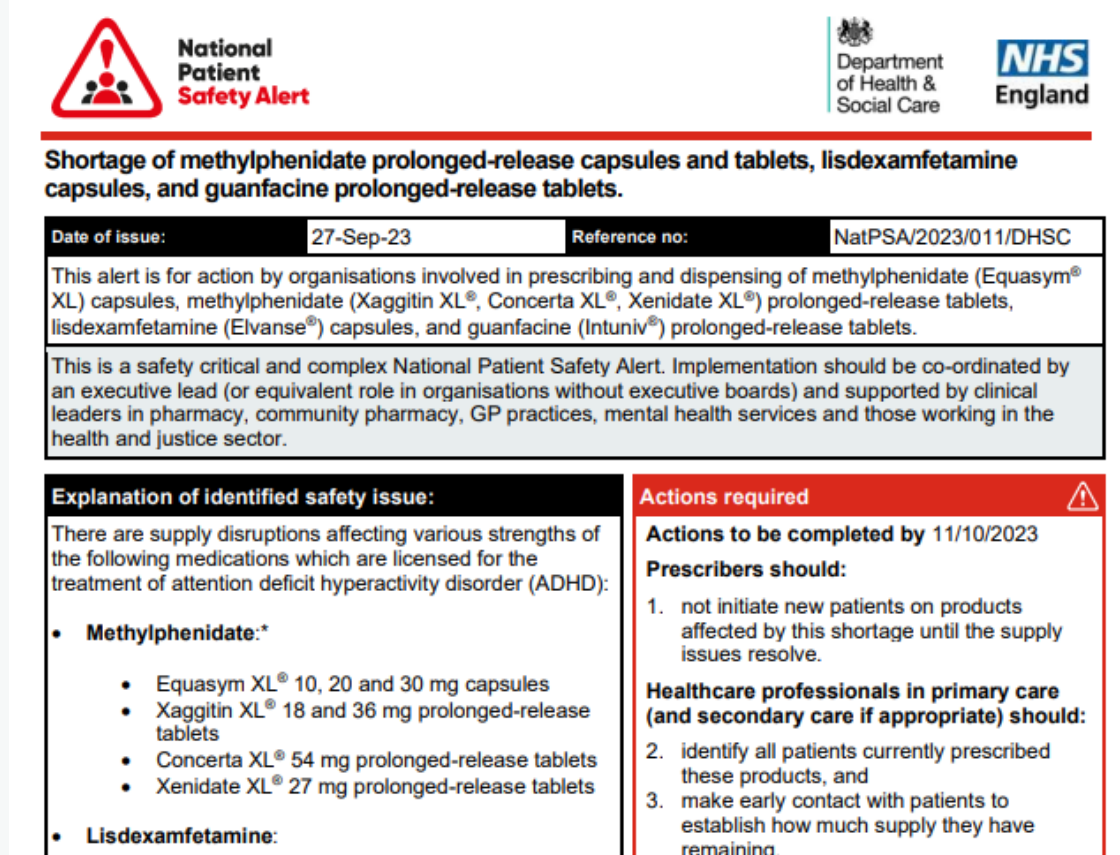
Figure 1: A schematic view of the dates of codes for a given patient. The colour corresponds to the month of code was recorded; yellow for ADHD diagnosis code, orange for Methylphenidate medication code and green for narcolepsy diagnosis code. The diagram shows it is impossible to determine which condition Methylphenidate was prescribed to.

Medication Caveat – Shortage

In September 2023, a National Patient Safety Alert was issued concerning shortage of ADHD medication.

We may caveat this external factor when interpreting the results on medication.

The statistics may have a bias for representation of patients accessing ADHD medication due to the shortage.



The screenshot displays a National Patient Safety Alert from the Department of Health & Social Care, NHS England. The alert is titled "Shortage of methylphenidate prolonged-release capsules and tablets, lisdexamfetamine capsules, and guanfacine prolonged-release tablets." It is dated 27-Sep-23 with reference number NatPSA/2023/011/DHSC. The alert text states that it is for action by organisations involved in prescribing and dispensing of methylphenidate (Equasym® XL) capsules, methylphenidate (Xaggitin XL®, Concerta XL®, Xenidate XL®) prolonged-release tablets, lisdexamfetamine (Elvanse®) capsules, and guanfacine (Intuniv®) prolonged-release tablets. It emphasizes that this is a safety critical and complex alert requiring co-ordination by an executive lead. The alert is structured into two main sections: "Explanation of identified safety issue" and "Actions required".

Date of issue:	27-Sep-23	Reference no:	NatPSA/2023/011/DHSC
This alert is for action by organisations involved in prescribing and dispensing of methylphenidate (Equasym® XL) capsules, methylphenidate (Xaggitin XL®, Concerta XL®, Xenidate XL®) prolonged-release tablets, lisdexamfetamine (Elvanse®) capsules, and guanfacine (Intuniv®) prolonged-release tablets.			
This is a safety critical and complex National Patient Safety Alert. Implementation should be co-ordinated by an executive lead (or equivalent role in organisations without executive boards) and supported by clinical leaders in pharmacy, community pharmacy, GP practices, mental health services and those working in the health and justice sector.			
Explanation of identified safety issue:	Actions required		
There are supply disruptions affecting various strengths of the following medications which are licensed for the treatment of attention deficit hyperactivity disorder (ADHD): • Methylphenidate:* • Equasym XL® 10, 20 and 30 mg capsules • Xaggitin XL® 18 and 36 mg prolonged-release tablets • Concerta XL® 54 mg prolonged-release tablets • Xenidate XL® 27 mg prolonged-release tablets • Lisdexamfetamine:	Actions to be completed by 11/10/2023 Prescribers should: 1. not initiate new patients on products affected by this shortage until the supply issues resolve. Healthcare professionals in primary care (and secondary care if appropriate) should: 2. identify all patients currently prescribed these products, and 3. make early contact with patients to establish how much supply they have remaining.		

Figure: Screenshot of Patient Safety Alert concerning the shortage of ADHD medication from the [Central Alerting System](#) on [behalf of the Medicines and Healthcare Products Regulatory Agency](#)

ADHD Medication – Long term perspective

We define cohort of patients that take ADHD medication for their ADHD

- 1. For each financial year, we consider patients that were registered for a least one day on a TPP provider.
- 2. We choose patients where the date of first ADHD medication is after their first diagnosis date.

Consider the following case studies:

- 1. Becky’s first ADHD diagnosis in 2020. She has been prescribed ADHD medication in 2021 and 2023.
- 2. Hans’ first ADHD diagnosis in 2023. He has been prescribed ADHD medication in 2021.

Year	Becky	Hans
2016		
2017		
2018		
2019		
2020		
2021		
2022		
2023		
2024		

Figure 1: A table showing how each case study would be counted (highlighted in blue) for each financial year.

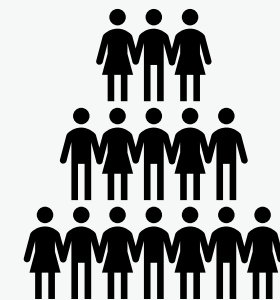
Medication – Short term perspective

We adapted the rules for the prescribing of antipsychotics in the ‘Health and Care of People with Learning Disabilities’ publication.

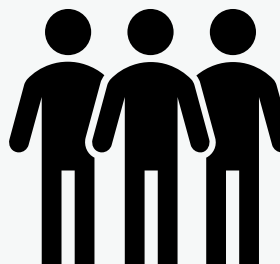
For a given calendar month, we compute two figures:

1. Let P_{dia} be patients that have an ADHD diagnosis code registered for the given month.
2. Let P_{med} be the subset of registered ADHD patients that have been prescribed ADHD medication in the last 6 months from the first day of interval month.

The final output is P_{med}/P_{dia}



Step 1: Find all patients that registered for at least one day in April 2025 and that have an ADHD diagnosis on or before April 30th 2025. This is our P_{dia}



Step 2: Filter the number of patients that have been prescribe at least one ADHD medication on or after 1st October 2024. This is our P_{med}

Figure 1: A diagrammatic example of computing the incidence of patient taking medication of the month of April 2025

Results

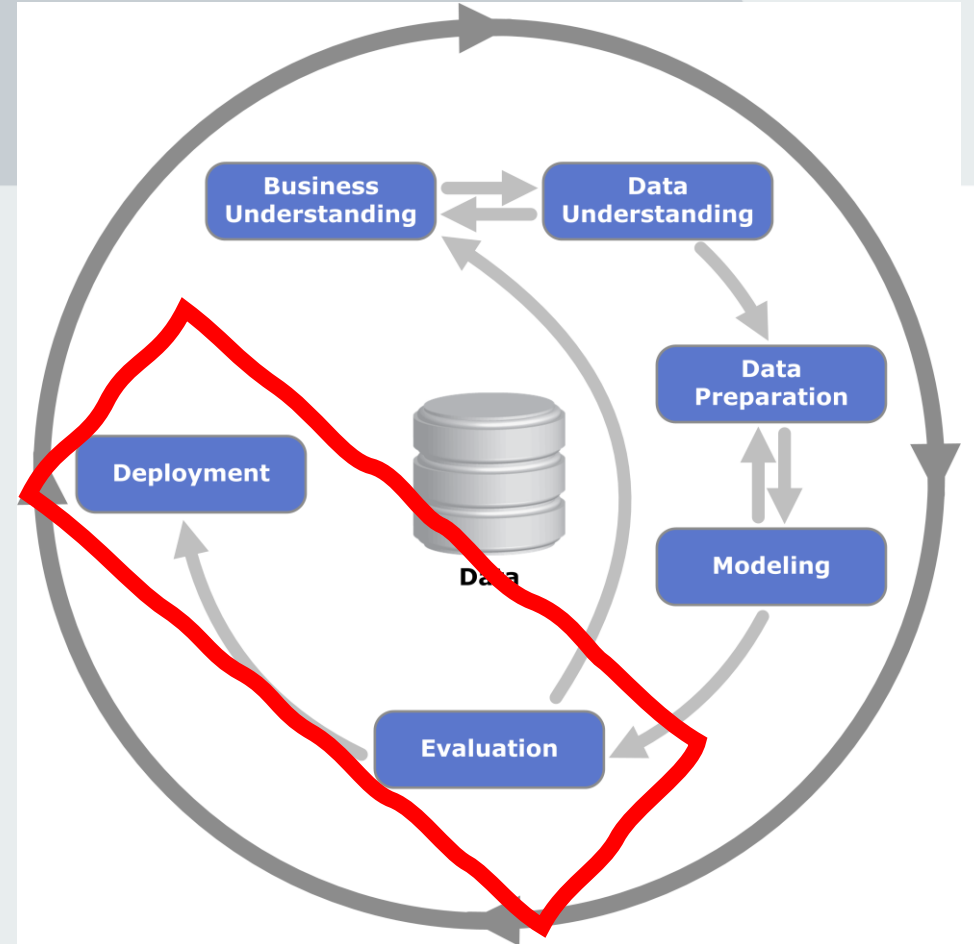
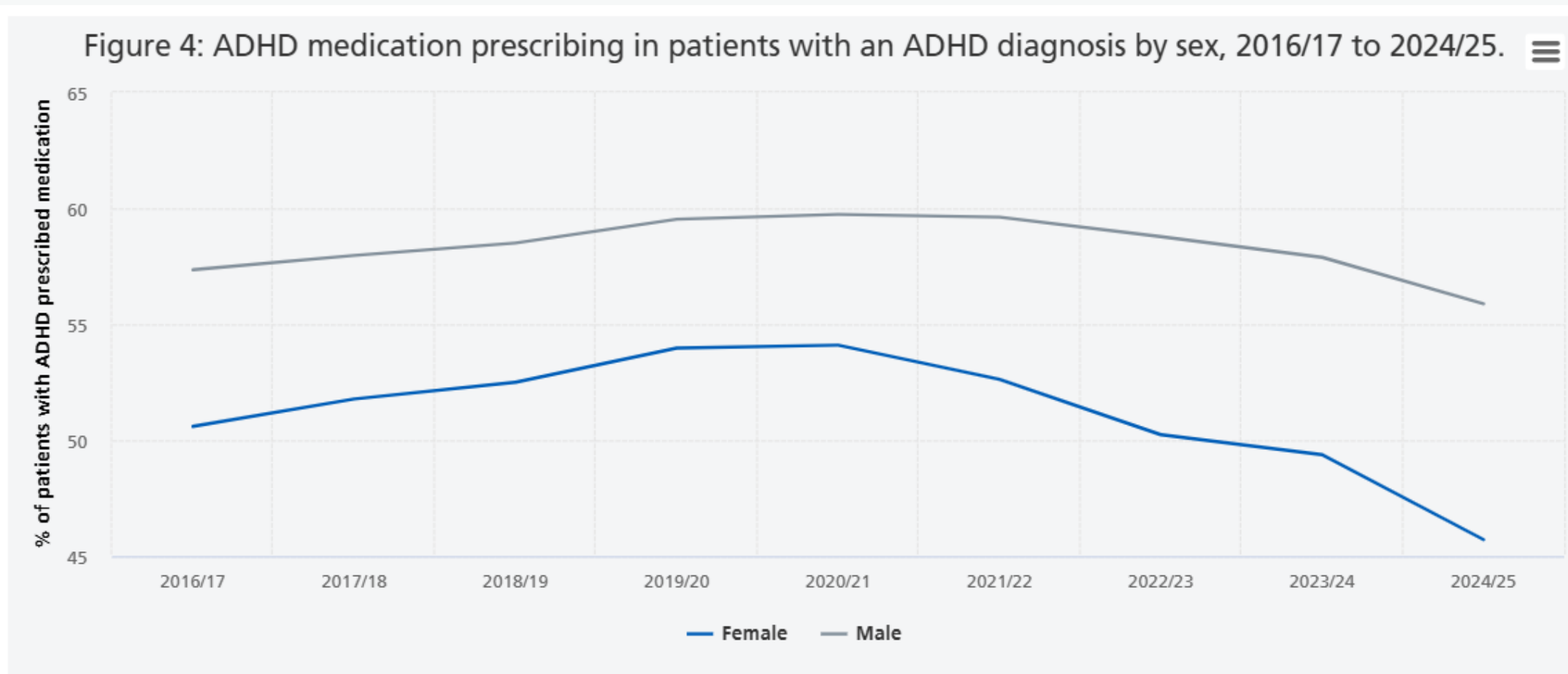


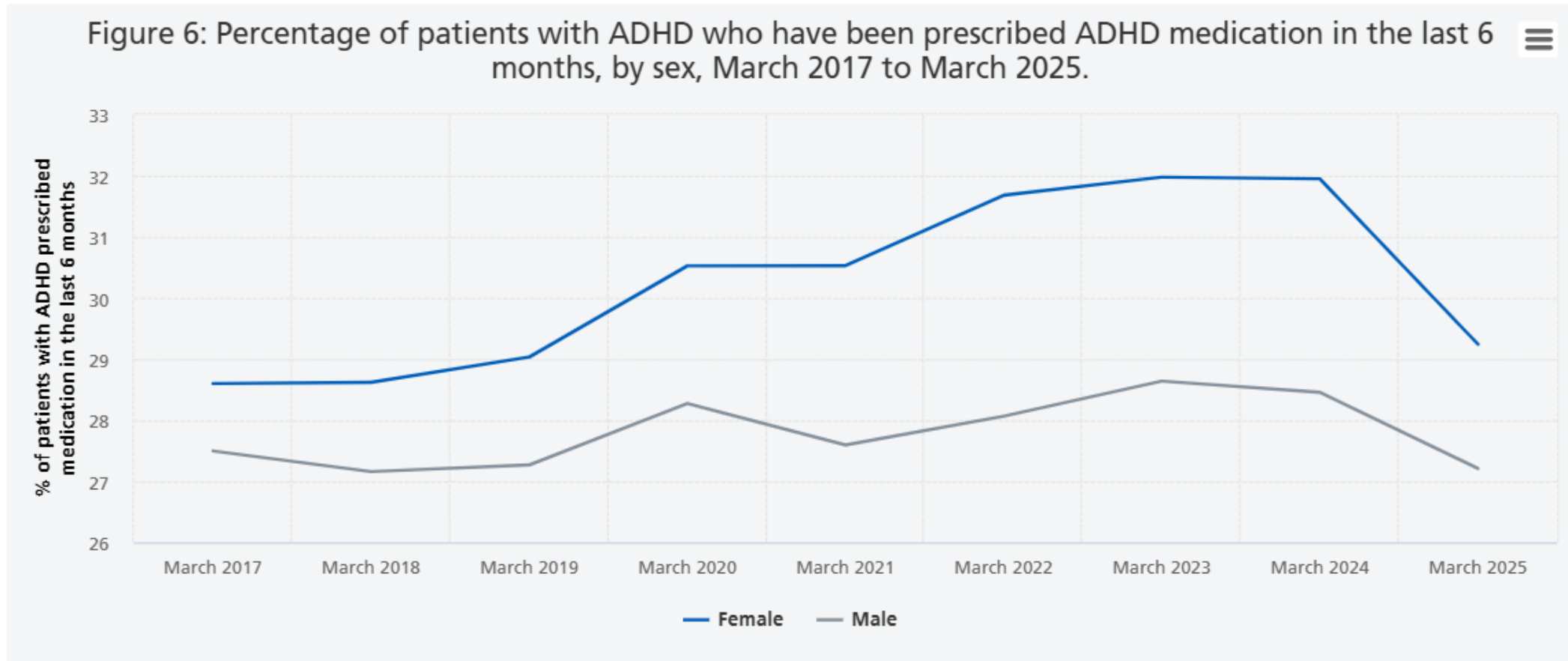
Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

Result Set 1: Medication (long term) by sex



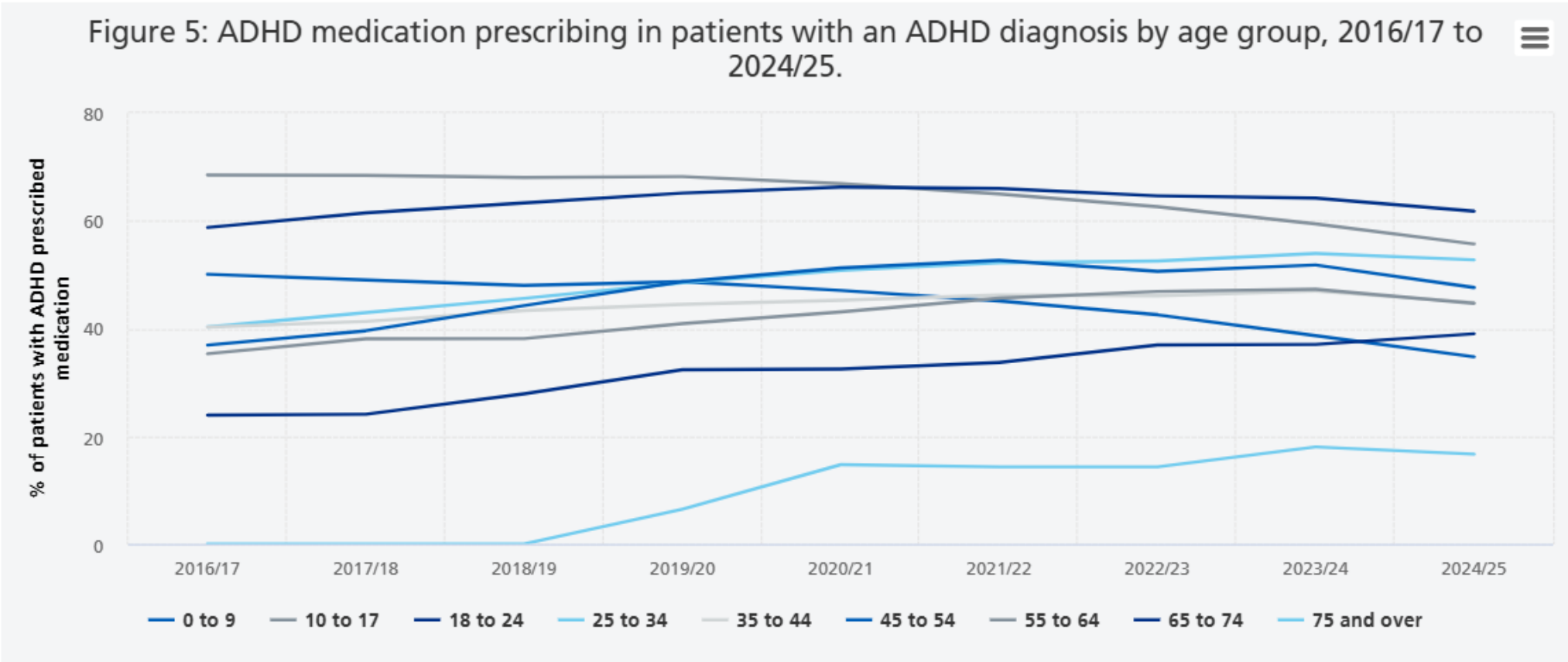
[Download the data for this chart Figure 4: ADHD medication prescribing in patients with an ADHD diagnosis by sex, 2016/17 to 2024/25.](#)

Result Set 1: Medication (short term) by sex



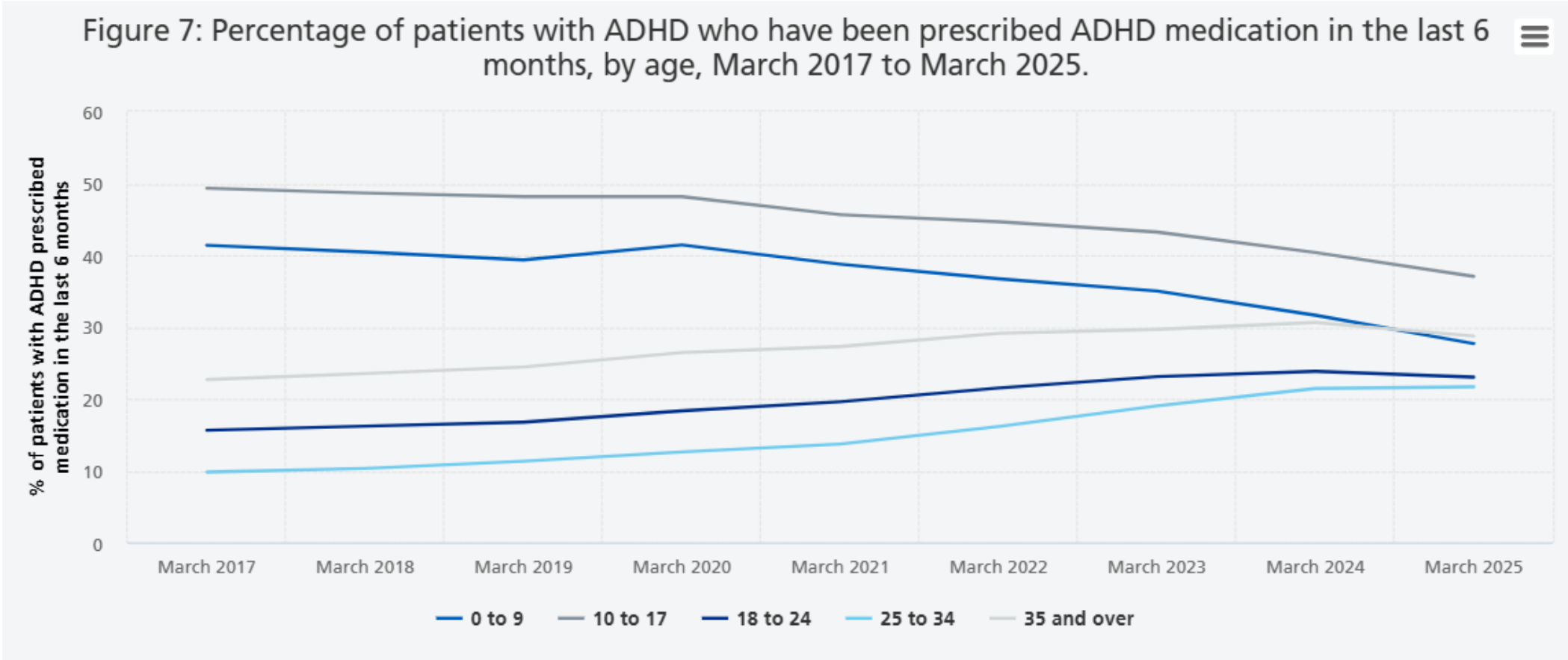
[Download the data for this chart Figure 6: Percentage of patients with ADHD who have been prescribed ADHD medication in the last 6 months, by sex, March 2017 to March 2025.](#)

Result Set 2: Medication (long term) by age band



[Download the data for this chart Figure 5: ADHD medication prescribing in patients with an ADHD diagnosis by age group, 2016/17 to 2024/25.](#)

Result Set 2: Medication (short term) by age band



[Download the data for this chart Figure 7: Percentage of patients with ADHD who have been prescribed ADHD medication in the last 6 months, by age, March 2017 to March 2025.](#)

Defining time between first ADHD diagnosis and medication

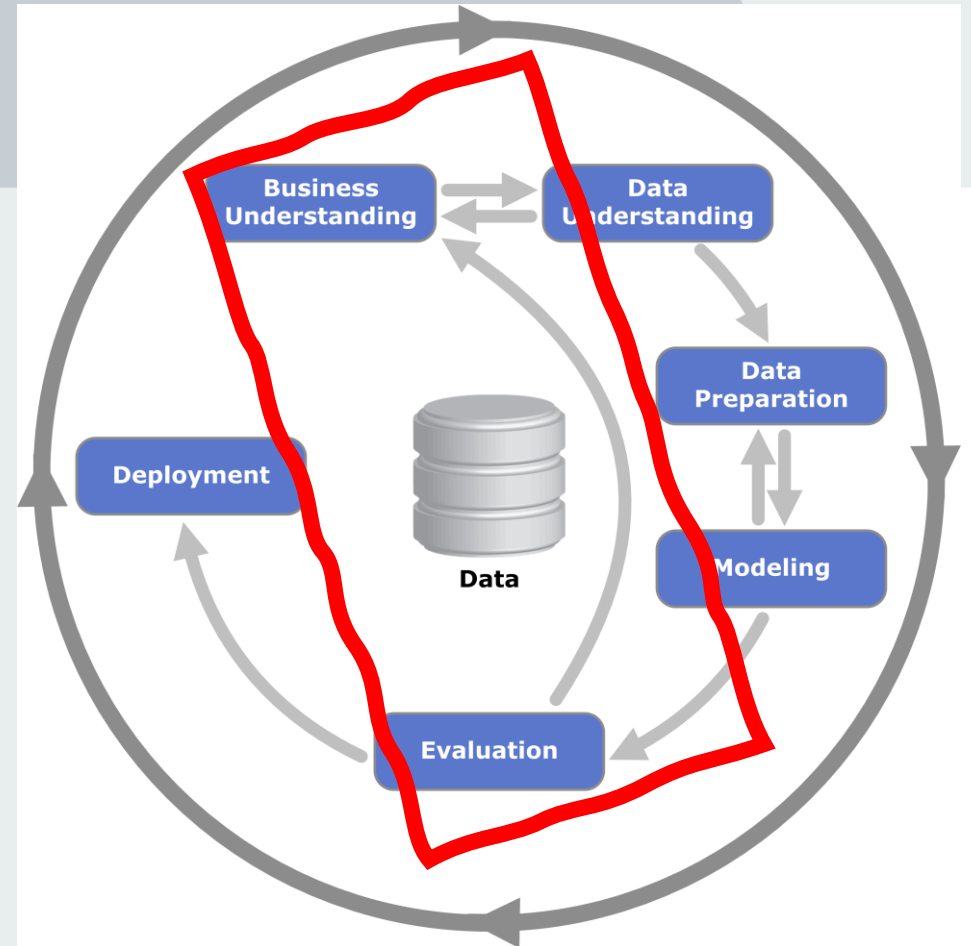


Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

Time between Diagnosis and Medication

We want to calculate the average time between first diagnosis and first medication.

Patients may be titrated and stabilised on ADHD medicines by NHS providers, including Right to Choose providers.

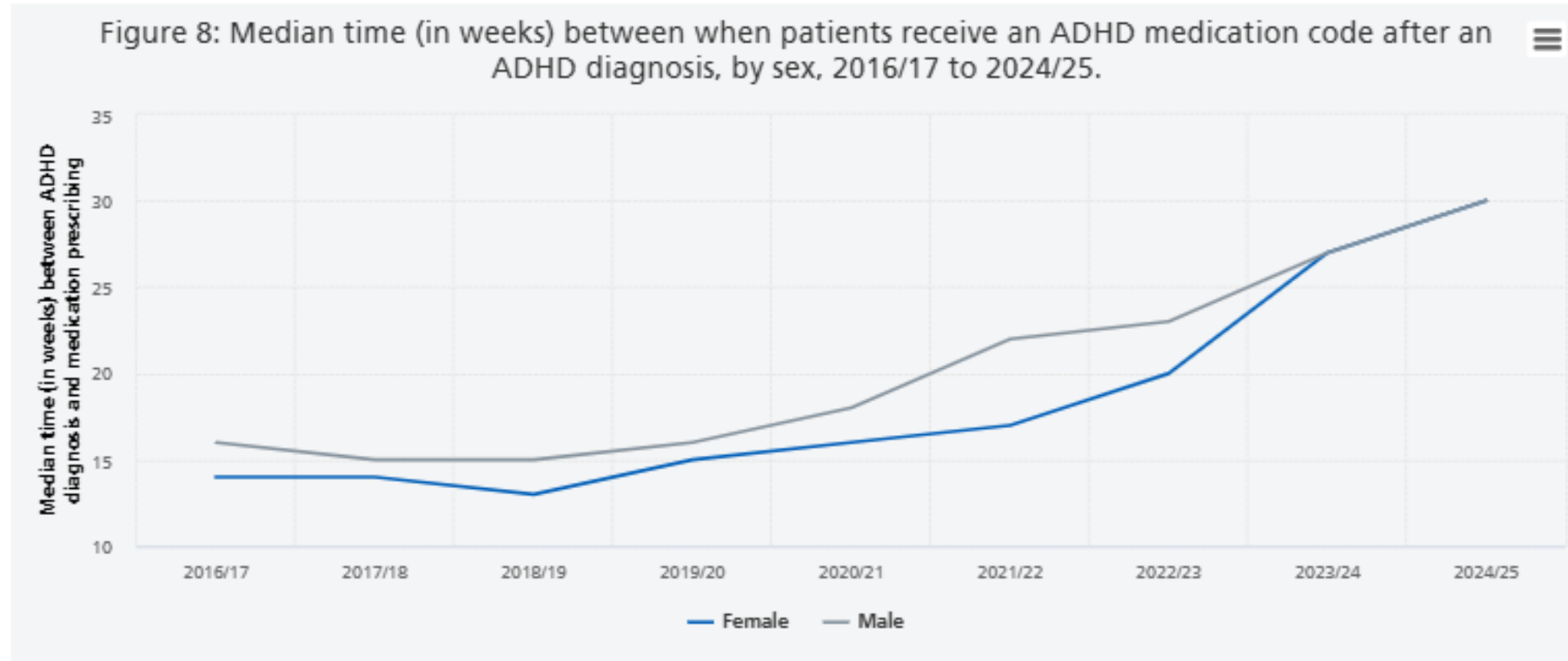
For a given financial year:

- 1. We cohort patients that have been prescribe their first ADHD medication within the given year
- 2. We trace back the respective patient's first ADHD diagnosis date
- 3. We compute the medium average time for each cohort in weeks.

Patient	Sam	Anna
First ADHD medication	01/03/2022	04/09/2023
Cohort Year	2021	2023
First ADHD diagnosis	22/12/2021	01/02/2023
Difference in weeks	10	31

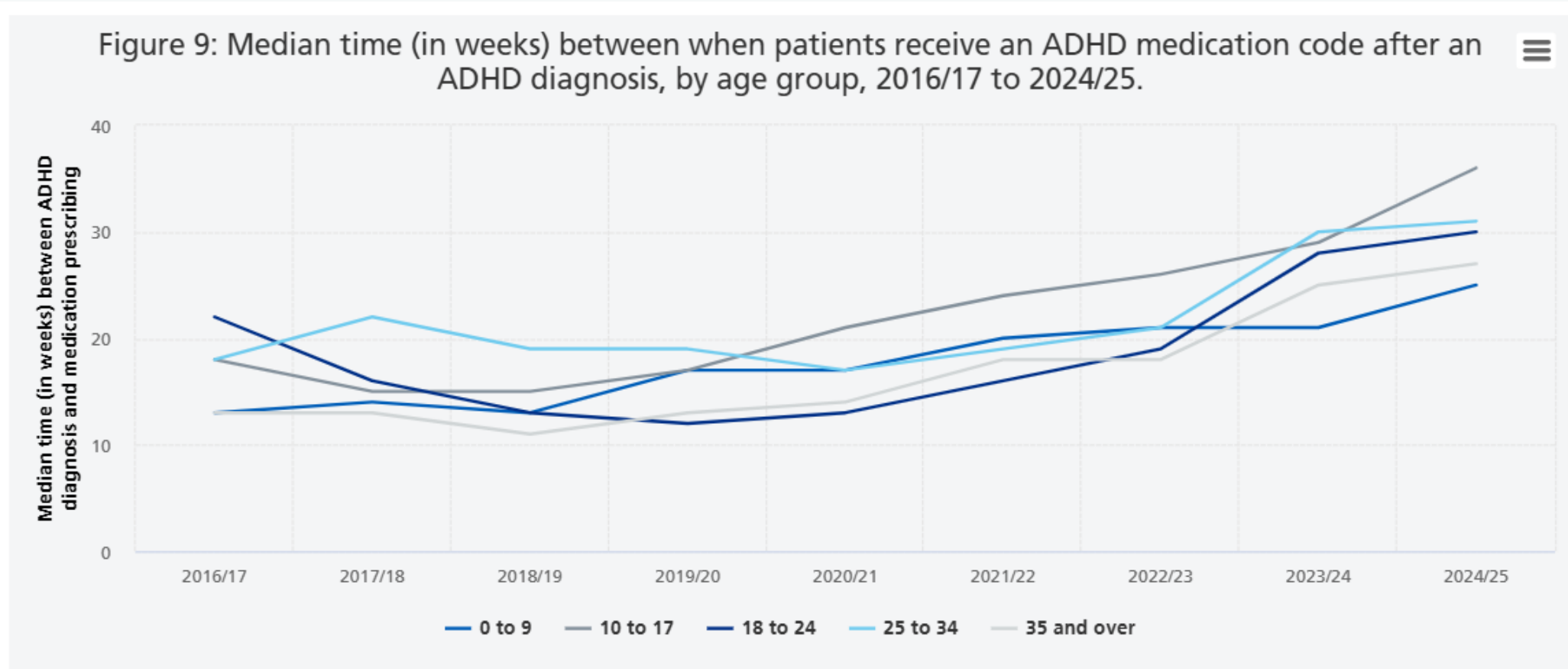
Figure 1: Case studies of how patients are slotted in each financial year. Patients can only be sorted in one financial year

Results: Time Diagnosis to Medication by sex



[Download the data for this chart Figure 8: Median time \(in weeks\) between when patients receive an ADHD medication code after an ADHD diagnosis, by sex, 2016/17 to 2024/25.](#)

Results: Time Diagnosis to Medication by age band



[Download the data for this chart Figure 9: Median time \(in weeks\) between when patients receive an ADHD medication code after an ADHD diagnosis, by age group, 2016/17 to 2024/25.](#)

Statistical Release



Figure: Steps in Cross-industry standard process for data mining adapted from [Wikipedia](#)

Closing Remarks

We produce statistics regarding ADHD diagnosis and medication.

Work was published as [official statistics](#) on NHS England.

This is the first time we have a computed the prevalence of the population with ADHD taking ADHD medication.

Output was shared with our stakeholders as a pre-release.

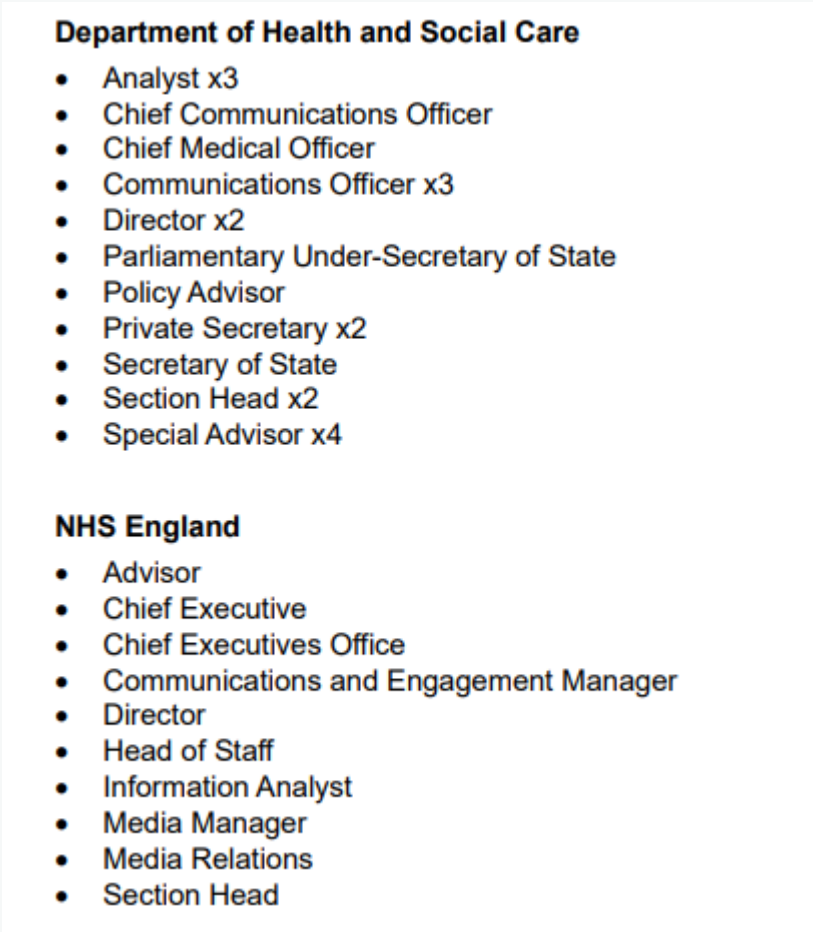


Figure 1: List of Organisations and Job Titles that was granted a pre-release for a period of 24 hours prior to the date set for publication (4th Dec 2025) under The Pre-release Access to Official Statistics Order 2008

Thank You



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